

Supplemental Note for Lecture 3

We can map the depth of any node in a randomized binary search tree (BST) to the number of comparisons involving that element during a corresponding execution of the Quicksort algorithm. Consequently, finding the expected average depth of a node in a BST reduces to finding the total expected number of comparisons performed by Quicksort. In the following derivation, we assume that all elements in the given array are equally likely to be chosen as the pivot.

Let C_n be the expected number of comparisons for an array of size n . We establish the following recurrence relation, where the first term represents the comparisons made at the current level of recursion, and the second term represents the expected number of comparisons across all possible partitions:

$$\begin{aligned} C_n &= (n-1) + \frac{1}{n} \sum_{i=0}^{n-1} (C_i + C_{n-i-1}) \\ &= (n-1) + \frac{2}{n} \sum_{i=0}^{n-1} C_i \end{aligned}$$

Multiplying both sides by n , we obtain:

$$nC_n = n(n-1) + 2 \sum_{i=0}^{n-1} C_i \tag{1}$$

Substituting $n-1$ for n yields:

$$(n-1)C_{n-1} = (n-1)(n-2) + 2 \sum_{i=0}^{n-2} C_i \tag{2}$$

Subtracting equation (2) from equation (1) gives:

$$\begin{aligned} nC_n - (n-1)C_{n-1} &= 2(n-1) + 2C_{n-1} \\ nC_n &= 2(n-1) + (n+1)C_{n-1} \end{aligned}$$

Dividing by $n(n+1)$ we get:

$$\begin{aligned}\frac{C_n}{n+1} &= \frac{C_{n-1}}{n} + \frac{2(n-1)}{n(n+1)} \\ &= \frac{C_{n-1}}{n} + \frac{2}{n+1} - \frac{2}{n(n+1)} \\ &\leq \frac{C_{n-1}}{n} + \frac{2}{n+1}\end{aligned}$$

By repeatedly applying this bound, we can expand the recurrence:

$$\begin{aligned}\frac{C_n}{n+1} &\leq \frac{C_{n-2}}{n-1} + 2\left(\frac{1}{n+1} + \frac{1}{n}\right) \\ &\vdots \\ \frac{C_n}{n+1} &\leq \frac{C_1}{2} + 2\left(\frac{1}{n+1} + \frac{1}{n} + \cdots + \frac{1}{3}\right)\end{aligned}$$

This bounded sum is related to the harmonic series, which we can approximate using an integral ($C_1 = 0$):

$$\frac{C_n}{n+1} \leq 2 \sum_{i=3}^{n+1} \frac{1}{i} \approx 2 \int_3^{n+1} \frac{1}{x} dx$$

Evaluating the integral yields our final asymptotic bounds:

$$\begin{aligned}\frac{C_n}{n+1} &\leq 2 \ln(n+1) \\ C_n &\leq 2(n+1) \ln(n+1) \\ C_n &= O(n \log n)\end{aligned}$$

Since the total expected number of comparisons is $O(n \log n)$, the expected average depth of a node is $\frac{C_n}{n} = O(\log n)$.

Note: It is also possible to prove that the probability of the maximum depth of a node (the height of the tree) exceeding logarithmic bounds is vanishingly small, though that proof requires a more involved probabilistic analysis.